Midterm 1 — Sample 1

DATA 412/612

Practice exam, questions only. Weeks 1–7: the Quiz 1 material, plus `rowwise()` and `across()`, `case_when()`, `readr`, exploratory plots, and `pivot_longer()`.

Put your answers in an R Markdown document and keep the code visible (`knitr::opts_chunk$set(echo = TRUE)` in the setup chunk). You may use your notes and the course website. You may not use Google, any other website, or an LLM. Do not discuss the exam with anyone else. Load the tidyverse. `mpg` comes with `ggplot2`.

1. **`case_when()`** (8 pts). In `mpg`, add `efficiency`: `"high"` when `hwy` is at least 30, `"mid"` when `hwy` is at least 20, and `"low"` otherwise. Then, for each `class` and `efficiency`, report the number of cars. Arrange so the largest counts come first.

2. **Row-wise and `across()`** (8 pts). Use this tibble.

```
quizzes <- tibble(
  student = c("Ada", "Lin", "Grace"),
  q1 = c(8, NA, 9),
  q2 = c(7, 10, NA),
  q3 = c(9, 8, 8)
)
```

   a. With `across()`, count the `NA`s in each quiz column.

   b. With `rowwise()` and `c_across()`, add each student's mean of `q1`, `q2`, and `q3`, ignoring missing quizzes.

3. **`readr` and a plot** (12 pts).

   a. Parse this frame. `when` should be a date, `price` a number (the dollar sign and the word `n/a` are not numbers), and `ok` logical, with the string `NA` meaning missing.

```
raw <- tibble(
  when  = c("2024-03-01", "2024-03-18", "2024-04-02", "2024-04-09"),
  price = c("$12.50", "8", "n/a", "$15"),
  ok    = c("TRUE", "FALSE", "NA", "TRUE")
)
```

   b. Using `mpg`, make one plot that compares highway mileage across `class`. State which class has the highest median `hwy`, and give that median.

4. **`pivot_longer()`** (12 pts). Reshape `wide` so there is one row per country and year, a numeric `year` column, and a `cases` column. Strip the `y` from the year names. Drop rows whose case count is missing. Then report the country and year with the most cases.

```
wide <- tibble(
  country = c("A", "B"),
  y2021 = c(10, NA),
  y2022 = c(12, 7),
  y2023 = c(9, 15)
)
```